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DESKTOP XY PLOTTER BASED PICK-AND-PLACE AND IN-CIRCUIT TEST SYSTEM FOR SMD RESISTORS AND DIODES

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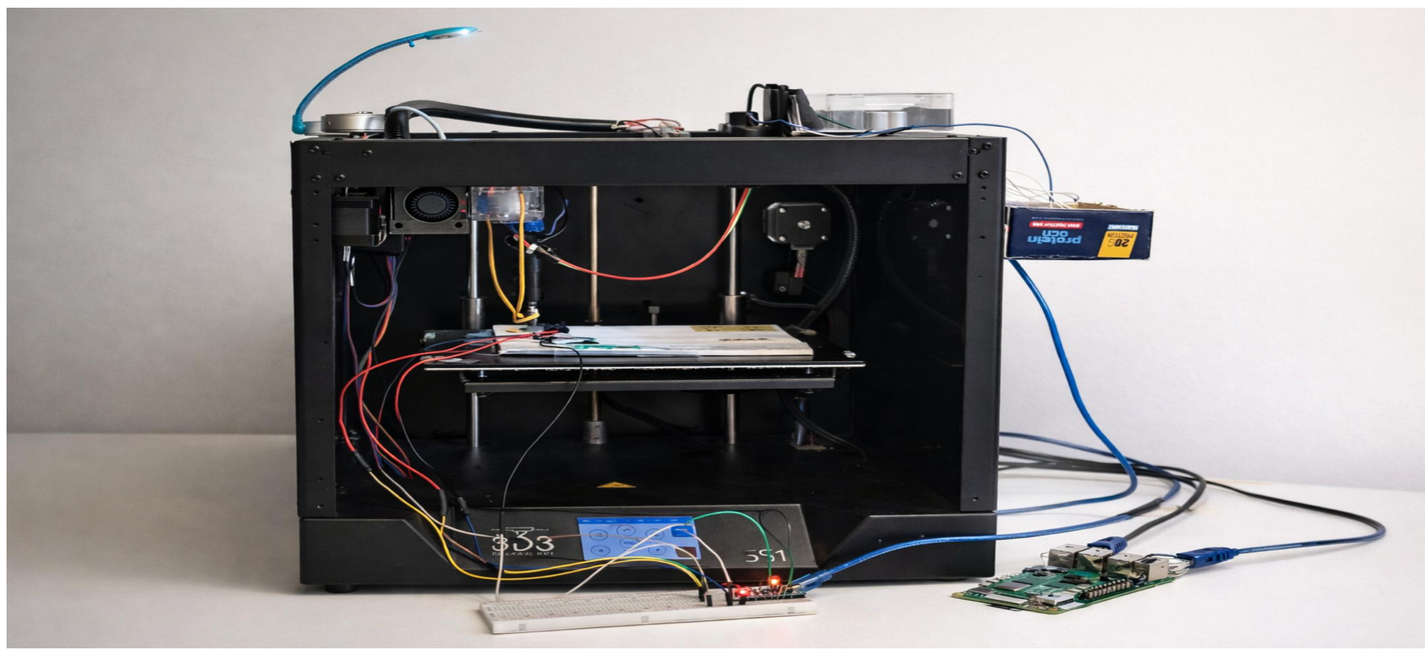
PROJECT OVERVIEW AND OBJECTIVES

This project realizes an autonomous mechatronic system that picks 2512 SMD resistors and DO-214AB rectifier diodes from a feeder area and places them onto a custom PCB with correct sequence, polarity, and repeatable positioning. The final prototype integrates motion control, vacuum pick-up, in-circuit electrical testing, computer vision, and a browser-based control interface in one compact platform.

The design addresses the ELE495 project scenario by combining autonomous component handling, optional pre-placement electrical testing, and real-time accuracy reporting on the user interface. High-level orchestration runs on Raspberry Pi, while embedded controllers handle deterministic actuation and test operations.

- Autonomous pick-and-place operation for two resistors and two diodes
- Vacuum-based non-magnetic picking mechanism
- Electrical testing for resistor value and diode polarity before placement
- Camera-based verification and live accuracy percentage reporting

FINAL PROTOTYPE



Final integrated prototype built on a repurposed desktop 3D-printer frame with vacuum pick-up, Raspberry Pi control, and fixed top camera.

SYSTEM ARCHITECTURE

The system follows a modular architecture that separates high-level decision making from low-level real-time hardware actions:

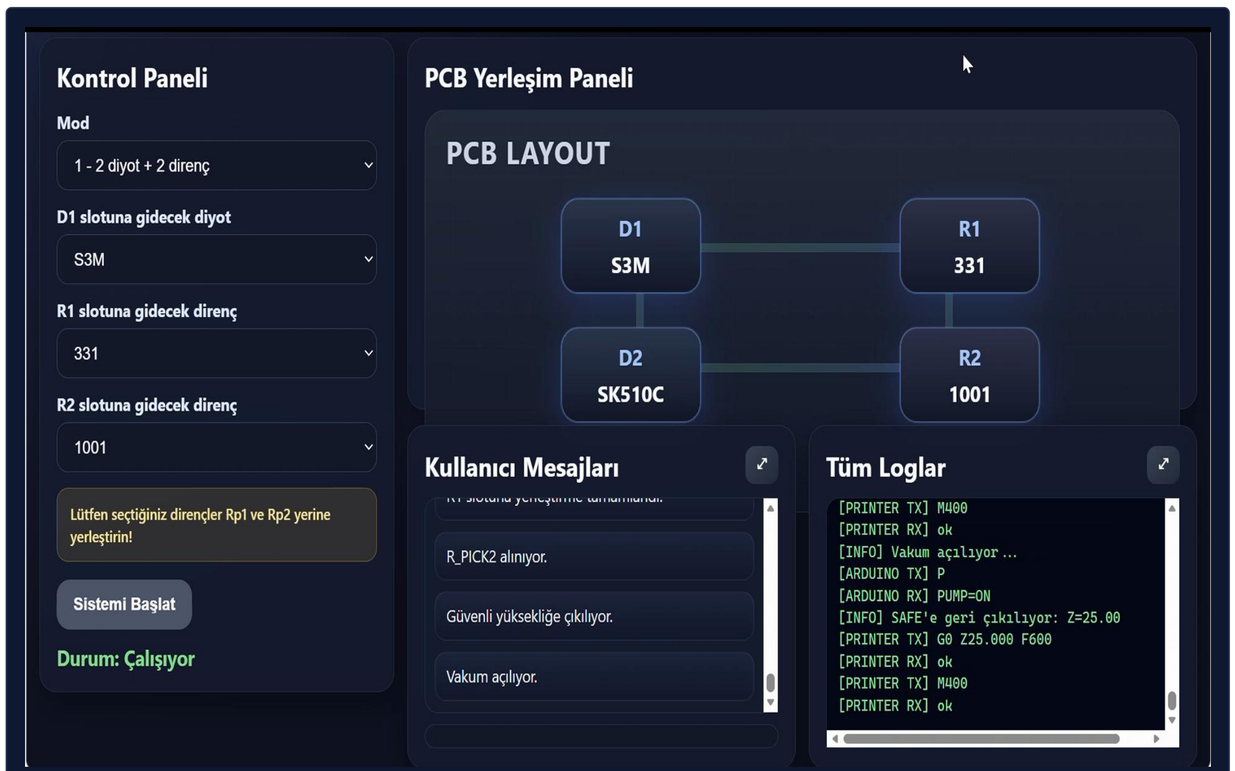
- Raspberry Pi 4: orchestration, OpenCV-based verification, serial communication, logging, and Flask web server
- Arduino Nano tester: resistor value detection and diode direction test before placement
- 3D-printer motion platform: Cartesian XY-Z movement using G-code based transport
- Vacuum tool and actuator electronics: stable pick, carry, release, and orientation correction support
- Fixed top USB camera: calibrated capture for post-placement inspection and scoring

OPERATIONAL WORKFLOW

A complete run combines mechanical transport, electrical test logic, user feedback, and computer-vision-based quality control:

1. Mode and slot selection through the web interface
2. Pickup from the feeder area using the vacuum nozzle
3. Electrical characterization in the test area
4. Placement onto the PCB footprint with controlled Z motion
5. Image capture from the fixed top camera
6. Per-slot and overall accuracy reporting on the UI

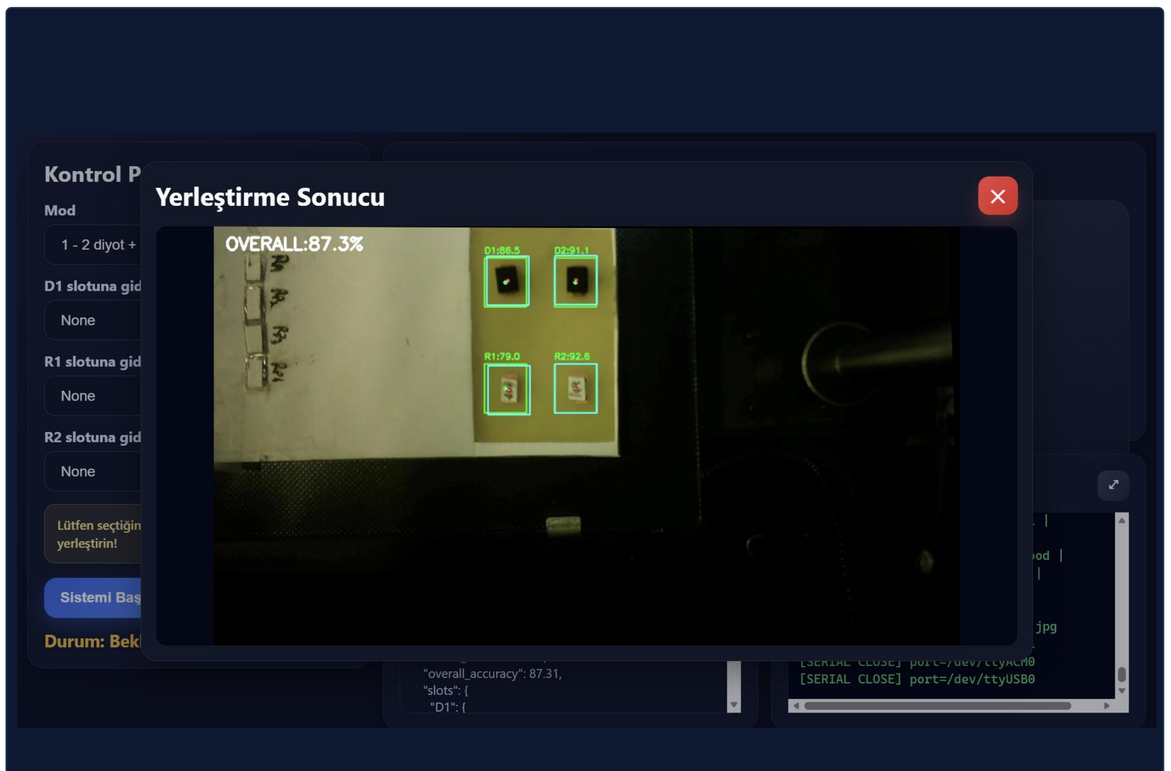
RASPBERRY PI WEB INTERFACE



Web-based operator panel with mode selection, PCB layout preview, user-friendly task feed, and detailed serial logs.

The Flask-based web interface was designed as the operator-facing layer of the system. It allows mode selection, slot configuration, start control, live status monitoring, user-level messages, and raw serial logs from all connected devices. Since the interface is hosted on the Raspberry Pi, it can be reached from any device on the same local network using a standard browser.

PLACEMENT VERIFICATION AND AOI



Camera-based placement verification window showing per-slot scoring and overall placement accuracy percentage.

For verification, the captured PCB image is warped into a fixed coordinate frame and compared with calibrated slot regions. The implemented Fixed-Coordinate IoU and Area-Ratio logic detects missing components, shifted placements, and orientation-related deviations. This transforms the final result into a measurable quality-control score rather than only a visual inspection.

RESULTS AND ACHIEVEMENTS

- The final prototype achieved full autonomous pick-and-place operation for both resistor and diode components.
- After the vacuum nozzle redesign, the system reached an approximately 95% successful pick rate.
- The bonus requirement of displaying placement accuracy percentages on the interface was completed.
- The optional electrical testing stage for resistor values and diode polarity was integrated before placement.
- Real-time status, test outputs, and verification logs were successfully delivered through the Web UI.

PROBLEMS ENCOUNTERED AND SOLUTIONS

- CSI camera socket failure on Raspberry Pi forced migration to a USB webcam and recalibration of the vision pipeline.
- Air leakage in the pneumatic pick head reduced handling reliability; this was solved by redesigning the nozzle and seal geometry.
- An overvoltage failure in the ESP32 test hardware led to migration of the testing subsystem to Arduino Nano.
- These changes confirmed the value of modular system partitioning and software adaptability.

ENGINEERING GAINS AND OUTLOOK

The project provided hands-on experience in mechatronic integration, embedded communication, machine vision, test instrumentation, and user-interface design. Beyond the course scope, the prototype demonstrates the feasibility of a low-cost desktop SMT assembly aid for laboratories, student teams, and small-scale prototyping environments. Future work can focus on higher repeatability, richer feeder mechanisms, faster cycle time, and broader component support.

Scan for GitHub Repository

Acknowledgements

We thank Dr. Zeki Uğurata Kocabıyıkoğlu for the project framework and evaluation criteria, and Coşku Kasnakoğlu for mentoring the design, integration, and final demonstration stages of the project.

References

[1] Final Report Group 3, ELE 495, 20 March 2026
[2] ELE 495 Project Criteria and course materials
[3] OpenPnP, LitePlacer, and vision-guided assembly references used in the reports